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1 A proof that π (and in fact π^2) is irrational

Theorem. π^2 (and therefore π) is irrational (As if π were rational π^2 would be too). Because rational numbers are exactly those whose decimals repeat (proven in Level 2), π 's decimals never repeat.

Proof. **Step 1.** We need to set everything up

Start by assuming $\pi^2 = \frac{a}{b}$, where a and b are positive integers. We aim to derive a contradiction. Define the sequence of integrals I_n as follows:

1. $I_0 = 2$
2. $I_1 = 4b$
3. $I_n = \frac{b^n}{n!} \int_0^\pi x^n (\pi - x)^n \sin(x) dx$ if $n \geq 2$

We can bound I_n as follows: It is non-negative since everything in the integral is non-negative and so is the coefficient, and $x^n (\pi - x)^n$ is bounded above by $\pi^n \pi^n = \pi^{2n}$ whenever $0 \leq x \leq \pi$, and therefore the integral (dropping the factor in front for now) can be bounded above by $\int_0^\pi \pi^{2n} \sin(x) dx = 2\pi^{2n}$. Therefore, if we consider I_n , we get

$$0 \leq I_n \leq 2 \frac{(\pi^2 b)^n}{n!}$$

Note that when, for example, $n > 2\pi^2 b$, the terms $2 \frac{(\pi^2 b)^n}{n!}$ will be at most half of the previous term (since the numerator will be multiplied by $\pi^2 b$ and the denominator by more than twice that), therefore these terms eventually tend to 0. Since I_n is bounded above by these terms and never negative, I_n also tends to 0.

The idea of the proof will be to show that if $\pi^2 = \frac{a}{b}$ then I_n is an integer for all n , which is a problem since I_n is always strictly between 0 and something that tends to 0 (and therefore is eventually less than 1), so I_n cannot be an integer when n is sufficiently large and this will be our contradiction.

Now we will do a classic A level further maths problem where we show from algebra that for n at least 2, $I_n = 2b(2n-1)I_{n-1} - abI_{n-2}$, which would mean that I_n is always an integer (since a , b , n , and I_0, I_1 are all integers) so we will be done.

Step 2. We need to verify what happens if n is 0 or 1. It turns out that the formula $I_n = \frac{b^n}{n!} \int_0^\pi x^n (\pi - x)^n \sin(x) dx$ holds for $n=0$ and $n=1$. The proof is as follows.

Case 1. For $n = 0$, the integral simplifies to

$$\frac{b^0}{0!} \int_0^\pi x^0 (\pi - x)^0 \sin(x) dx = 2$$

Since $0! = 1$ (If I haven't mentioned this, it's because $2! = \frac{3!}{3}$, $1! = \frac{2!}{2}$, and $0! = \frac{1!}{1} = 1$)

Case 2. For $n = 1$, the integral simplifies to

$$b \int_0^\pi x (\pi - x) \sin(x) dx = \pi b \int_0^\pi x \sin(x) dx - b \int_0^\pi x^2 \sin(x) dx$$

Integration by parts on the first integral setting

$$u = x, \frac{dv}{dx} = \sin(x) \text{ so } \frac{du}{dx} = 1, v = -\cos(x)$$

gives that it is exactly equal to

$$-[x \cos(x)]_0^\pi - \int_0^\pi -\cos(x)dx = -(-\pi - 0) - 0 = \pi$$

Also, $\int_0^\pi x^2 \sin(x)dx$ can be integrated by parts if we set

$$u = x^2, dv = \sin(x)dx \text{ so } du = 2x dx, v = -\cos(x)$$

meaning that the integral is then equal to

$$-[x^2 \cos(x)]_0^\pi - \int_0^\pi -2x \cos(x)dx = -(\pi^2 - 0) + \int_0^\pi 2x \cos(x)dx = \pi^2 + \int_0^\pi 2x \cos(x)dx$$

Integrating the second integral by parts, setting

$$u = 2x, \frac{dv}{dx} = \cos(x) \text{ so } \frac{du}{dx} = 2, v = \sin(x)$$

we get that

$$\int_0^\pi 2x \cos(x)dx = [2x \sin(x)]_0^\pi - \int_0^\pi 2 \sin(x)dx = (0 - 0) - 4 = -4$$

Therefore we conclude

$$\int_0^\pi x^2 \sin(x)dx = \pi^2 - 4$$

Now $I_1 = b\pi(\pi) - b(\pi^2 - 4) = 4b$, as required.

Step 3. We will prove the recurrence relation $I_n = 2b(2n - 1)I_{n-1} - abI_{n-2}$ for n at least 2. As discussed, this would establish I_n is always an integer but it is positive and tends to 0 which would give a contradiction. This is not conceptually difficult, just a very long calculus exercise.

Consider $\int_0^\pi x^n(\pi - x)^n \sin(x)dx$. We will do this by parts. Integrating by parts twice will reduce the powers, allowing us to express the integral using the two preceding ones. Set

$$u = x^n(\pi - x)^n, \frac{dv}{dx} = \sin(x)$$

By the product and chain rules,

$$\frac{du}{dx} = nx^{n-1}(\pi - x)^n - nx^n(\pi - x)^{n-1} = x^{n-1}(\pi - x)^{n-1}[n(\pi - x) - nx] = nx^{n-1}(\pi - x)^{n-1}(\pi - 2x)$$

but u vanishes (is 0) at 0 and π and therefore the $[uv]_0^\pi$ term from the integration by parts formula does too. Therefore the integration by parts formula gives (since $v = -\cos(x)$)

$$\int_0^\pi x^n(\pi - x)^n \sin(x)dx = n \int_0^\pi x^{n-1}(\pi - x)^{n-1}(\pi - 2x) \cos(x)dx$$

Now we want to integrate this by parts again so we need to work out the derivative of $x^{n-1}(\pi - x)^{n-1}(\pi - 2x)$. By the product rule on the last term, this is equal to

$$-2x^{n-1}(\pi - x)^{n-1} + (\pi - 2x) \frac{d}{dx} [x^{n-1}(\pi - x)^{n-1}]$$

but this last derivative is just the $n - 1$ version of something whose derivative we already know. Therefore the original derivative we wanted i.e. the derivative of $x^{n-1}(\pi - x)^{n-1}(\pi - 2x)$ is

$$-2x^{n-1}(\pi - x)^{n-1} + (\pi - 2x)^2(n - 1)x^{n-2}(\pi - x)^{n-2}$$

Therefore for the integral

$$\int_0^\pi x^{n-1}(\pi - x)^{n-1}(\pi - 2x) \cos(x)dx$$

we set

$$u = x^{n-1}(\pi - x)^{n-1}(\pi - 2x), \frac{dv}{dx} = \cos(x)$$

so now we have

$$v = \sin(x), \frac{du}{dx} = -2x^{n-1}(\pi - x)^{n-1} + (\pi - 2x)^2(n-1)x^{n-2}(\pi - x)^{n-2}$$

This time v vanishes at 0 and π so again the uv term vanishes and we reduce to

$$-\int_0^\pi v \frac{du}{dx} dx = \int_0^\pi \left[2x^{n-1}(\pi - x)^{n-1} - (\pi - 2x)^2(n-1)x^{n-2}(\pi - x)^{n-2} \right] \sin(x) dx$$

By some algebra, this is

$$\int_0^\pi \left[2x^{n-1}(\pi - x)^{n-1} - (\pi^2 - 4x(\pi - x))(n-1)x^{n-2}(\pi - x)^{n-2} \right] \sin(x) dx$$

This is therefore equal to

$$\int_0^\pi \left[(2 + 4(n-1))x^{n-1}(\pi - x)^{n-1} - (\pi^2)(n-1)x^{n-2}(\pi - x)^{n-2} \right] \sin(x) dx$$

Putting all this together

$$\begin{aligned} I_n &= \frac{b^n}{n!} \int_0^\pi x^n (\pi - x)^n \sin(x) dx = \frac{b^n}{n!} n \int_0^\pi x^{n-1} (\pi - x)^{n-1} (\pi - 2x) \cos(x) dx \\ &= \frac{b^n}{(n-1)!} \left[\int_0^\pi \left[2 + 4(n-1) \right] x^{n-1} (\pi - x)^{n-1} - (\pi^2)(n-1)x^{n-2}(\pi - x)^{n-2} \right] \sin(x) dx \\ &= b \frac{b^{n-1}}{(n-1)!} \int_0^\pi \left[(2 + 4(n-1))x^{n-1}(\pi - x)^{n-1} \right] \sin(x) dx - \frac{b^2 \pi^2}{n-1} \frac{b^{n-2}}{(n-2)!} \left[\int_0^\pi \left[(n-1)x^{n-2}(\pi - x)^{n-2} \right] \sin(x) dx \right] \\ &= b I_{n-1} (2 + 4(n-1)) - b^2 \pi^2 \frac{b^{n-2}}{(n-2)!} \left[\int_0^\pi \left[x^{n-2}(\pi - x)^{n-2} \right] \sin(x) dx \right] \\ &= 2b(2n-1) I_{n-1} - ab I_{n-2} \end{aligned}$$

where we have used the fact that $\pi^2 = \frac{a}{b}$ (We had to use it at some point otherwise we know something has gone wrong). This completes the proof of step 3.

□

2 A proof that $\sum_{n=1}^\infty \frac{1}{n^2} = \frac{\pi^2}{6}$

Theorem. $\sum_{n=1}^\infty \frac{1}{n^2} = \frac{\pi^2}{6}$

Proof. Recall De Moivre's theorem, which says that $(\cos(\theta) + i\sin(\theta))^n = \cos(n\theta) + i\sin(n\theta)$. Assume that n is a positive integer and $\sin(\theta) \neq 0$. Now we have that

$$\begin{aligned} \frac{(\cot(\theta) + i)^{2n+1} - (\cot(\theta) - i)^{2n+1}}{2i} &= \frac{(\cos(\theta) + i\sin(\theta))^{2n+1} - (\cos(\theta) - i\sin(\theta))^{2n+1}}{2i(\sin^{2n+1}(\theta))} \\ &= \frac{\cos((2n+1)\theta) + i\sin((2n+1)\theta) - \cos((2n+1)\theta) + i\sin((2n+1)\theta)}{2i(\sin^{2n+1}(\theta))} = \frac{\sin((2n+1)\theta)}{\sin^{2n+1}(\theta)}. \end{aligned}$$

Now let's suppose that $\theta = \frac{m\pi}{2n+1}$ for some integer m that is between 1 and n inclusive. Then we know that $\sin((2n+1)\theta) = 0$ and therefore $\frac{(\cot(\theta)+i)^{2n+1} - (\cot(\theta)-i)^{2n+1}}{2i} = 0$. Now I will use the binomial theorem to expand $(\cot(\theta) + i)^{2n+1} - (\cot(\theta) - i)^{2n+1}$, which we know must be equal to 0. We will get

$$0 = \sum_{r=0}^{2n+1} \binom{2n+1}{r} \cot^r(\theta) i^{2n+1-r} - \sum_{r=0}^{2n+1} \binom{2n+1}{r} \cot^r(\theta) (-i)^{2n+1-r} = \sum_{r=0}^{2n+1} \binom{2n+1}{r} \cot^r(\theta) \left[i^{2n+1-r} - (-i)^{2n+1-r} \right].$$

When r is odd, $2n + 1 - r$ is even, so the bracketed term will look like $i^{2k} - (-i)^{2k}$ where k is an integer, but this equals $(-1)^k - (-1)^k = 0$. Therefore the only terms that survive are the terms where r is even, so we can write $r = 2s$, where s runs from 0 to n . We have

$$\sum_{s=0}^n \binom{2n+1}{2s} \cot^{2s}(\theta) \left[i^{2(n-s)+1} - (-i)^{2(n-s)+1} \right]$$

Let's replace s with $n-s$, and note the identity $\binom{2n+1}{2s} = \binom{2n+1}{2n+1-2s}$ (since we can choose $2s$ things in the same number of ways we can exclude $2s$ things). Set $k = n - s$, then we have

$$\sum_{k=0}^n \binom{2n+1}{2k+1} \cot^{2(n-k)}(\theta) \left[i^{2k+1} - (-i)^{2k+1} \right]$$

This last term is $i + i = 2i$ when k is even and $-i - i = -2i$ when k is odd. Since the whole thing equals 0, we can divide it through by $2i$ to get the following:

$$\sum_{k=0}^n \binom{2n+1}{2k+1} \cot^{2(n-k)}(\theta) (-1)^k = 0$$

If $x = \cot^2(\theta)$ we can write the sum above as

$$\binom{2n+1}{1} x^n - \binom{2n+1}{3} x^{n-1} + \dots + (-1)^n = 0$$

Note that the roots of the polynomial above are exactly the n numbers $\cot^2\left(\frac{m\pi}{2n+1}\right)$ for m an integer going from 1 to n . The same logic applies if m is an integer greater than n or less than 0 and generally any non-multiple of $2n+1$, however this would give one of the same numbers – this can be argued from symmetry of the graph, or from the fact that it must be the case as $\cot^2\left(\frac{m\pi}{2n+1}\right)$ for m from 1 to n is always distinct as $\cot$ is strictly decreasing in that range, and an n -degree polynomial cannot have more than n roots (or else you could factor it more than n times by the factor theorem). Cotangent is positive on this interval so squaring preserves distinctness.

Now we know that the sum of roots of a polynomial is minus the second leading coefficient divided by the leading coefficient. Applying this to the polynomial above gives

$$\sum_{m=1}^n \cot^2\left(\frac{m\pi}{2n+1}\right) = \frac{\binom{2n+1}{3}}{\binom{2n+1}{1}} = \frac{\frac{(2n+1)!}{3!(2n-2)!}}{2n+1} = \frac{(2n)!}{6(2n-2)!} = \frac{(2n)(2n-1)}{6} = \frac{n(2n-1)}{3}$$

We now need two facts:

1. For θ between 0 and $\frac{\pi}{2}$, $\sin(\theta) = \int_0^\theta \cos(x) dx$. Since $\cos(x)$ is always between 0 and 1 in that range, it means that $\sin(\theta)$ is between 0 and θ
2. For θ between 0 and $\frac{\pi}{2}$, $\tan(\theta) = \int_0^\theta \sec^2(x) dx$ (you can write $\tan$ as $\frac{\sin}{\cos}$ and differentiate it using the quotient rule and verify that it is $\sec^2$). Since $\cos(x)$ is always between 0 and 1 in that range, it means that $\sec(\theta)$ is greater than 1 for all non-zero values in that range, and therefore $\tan(\theta) > \theta$

So $0 < \sin(\theta) < \theta < \tan(\theta)$ for $0 < \theta < \frac{\pi}{2}$. Since in $\sin(\theta) < \theta < \tan(\theta)$ they are all positive, taking the reciprocal of both sides flips the inequality, so $\cot(\theta) < \frac{1}{\theta} < \csc(\theta)$. We still have positive things, so we will square both sides to get $\cot^2(\theta) < \frac{1}{\theta^2} < \csc^2(\theta)$. However we can say something else, which is that $\cot^2(\theta) < \frac{1}{\theta^2} < 1 + \cot^2(\theta)$, as

$$\csc^2(\theta) = \frac{1}{\sin^2(\theta)} = \frac{\sin^2(\theta) + \cos^2(\theta)}{\sin^2(\theta)} = 1 + \cot^2(\theta)$$

Now

$$\sum_{m=1}^n \cot^2\left(\frac{m\pi}{2n+1}\right) < \sum_{m=1}^n \frac{(2n+1)^2}{m^2 \pi^2} < \sum_{m=1}^n 1 + \cot^2\left(\frac{m\pi}{2n+1}\right) = n + \sum_{m=1}^n \cot^2\left(\frac{m\pi}{2n+1}\right)$$

But we know

$$\sum_{m=1}^n \cot^2 \left(\frac{m\pi}{2n+1} \right) = \frac{n(2n-1)}{3}$$

Therefore

$$n + \sum_{m=1}^n \cot^2 \left(\frac{m\pi}{2n+1} \right) = \frac{n(2n+2)}{3}$$

Therefore we have the tight bound

$$\frac{n(2n-1)}{3} < \sum_{m=1}^n \frac{(2n+1)^2}{m^2 \pi^2} < \frac{n(2n+2)}{3}$$

for all positive integers n. Let's multiply through by $\frac{\pi^2}{(2n+1)^2}$ to get that

$$\frac{n(2n-1)\pi^2}{3(2n+1)^2} < \sum_{m=1}^n \frac{1}{m^2} < \frac{n(2n+2)\pi^2}{3(2n+1)^2}$$

But notice that as n gets large, $\frac{n(2n-1)}{3(2n+1)^2}$ and $\frac{n(2n+2)}{3(2n+1)^2}$ will both approach $\frac{1}{6}$, as they are exactly $\frac{1}{6}$ save for the +1 and +2 and -1 terms which contribute a factor on the order of $\frac{1}{n}$. Therefore the sum $\sum_{m=1}^n \frac{1}{m^2}$ is bounded below and above by something that approaches $\frac{\pi^2}{6}$, and therefore the infinite sum is equal to exactly $\frac{\pi^2}{6}$.

□